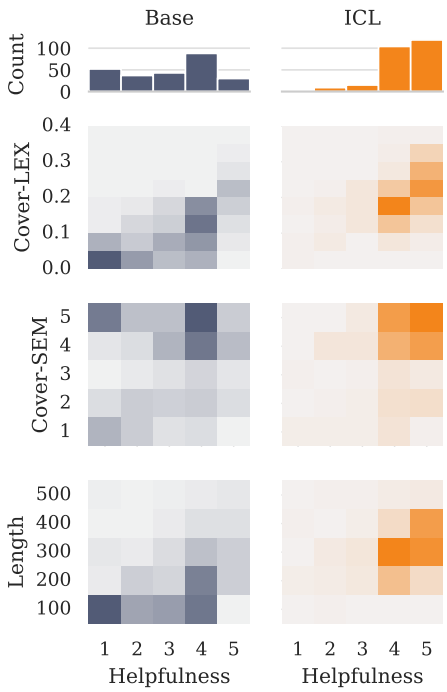

Figure 2: The relationship between lexical coverage, semantic coverage, response length, and helpfulness (x-axis) in LIMA-OE. Cover-LEX and Cover-SEM are computed with respect to Llama 2 Chat, with smaller values corresponding to more content missing from the aligned model response that the base model surfaces. When there is less overlap between the base and aligned model, base response responses are of lower quality. Helpful responses from the base model tend to cover the same content as the reference under both coverage metrics.

- Cover-LEX computes the fraction of words shared between responses (Jaccard similarity): $|\mathbf{y} \cap \mathbf{y}_r|/|\mathbf{y} \cup \mathbf{y}_r|$. This metric is identical to the Jaccard similarity from Section 2.3, but computed between responses from different models on the same input.
- Cover-SEM: We measure semantic coverage by prompting GPT-4 to assess if $\mathbf{y_r}$ covers all information in $\mathbf{y}$. Each output pair is rated on a scale of 1 to 5 where 1 means "$\mathbf{y}$ contains completely new information" and a 5 means "$\mathbf{y}$ is redundant given $\mathbf{y_r}$."

For both metrics, lower numbers signify there is content in the base model response that is not present in the aligned model response.

We also evaluate the helpfulness of the base response. We use Prompt C.4 to assign a score on a scale from 1 to 5.

We use Llama 2 Chat as the reference model and Llama 2 Base and Llama 2 ICL as base model responses. As we aim to find useful information missing from reference model responses, we draw five samples from the base model for each reference model response.

Results Figure 2 presents results showing, for each response pair, the relationship between coverage and helpfulness. When coverage metrics are low (i.e., the base response contains new information), the quality of the base response also tends to be low. For instance, in Figure 2 (Cover-SEM), the bottom-left purple cell represents points that are not covered semantically, but which are also not deemed to be helpful. ICL responses are generally higher quality than base responses, but this also manifests as a comparatively higher amount of coverage. In conclusion, we find ICL alignment induces increased distributional similarities between models, and the aligned model does not lose much high-quality information.

3.2 Pluralistic stances post-alignment

If alignment reduces lexical diversity, could it cause models to suppress certain viewpoints and reduce diversity along pluralistic dimensions as well? We explore this issue by appealing to the binary nature of questions in CONFLICTINGQA and examining how response stances change post-alignment.

Setting We analyze stances (AlDayel and Magdy, 2021) for Llama 2 Base, Llama 2 Chat, and Llama 2 ICL for each question in CONFLICTINGQA and $N = 5$ samples per question. Each response is classified as one of "yes", "no", "both", or "other", with "both" corresponding to responses that present both perspectives and "other" corresponding to responses that are off-topic or do not answer the question. We classify stances by prompting GPT-4; details of the prompt are given in Appendix C.

Results Figure 3a presents the stance distribution for each model. We observe a marked shift toward more pluralistic "both" responses in aligned LLMs. On the other hand, a relatively small number of Llama 2 Base responses are classified as "both", with the majority being "yes" or "no".

We also analyzed the homogeneity of stances to the same question as measured by entropy. There are only six possible unique entropy values and we depict each separately along an ordinal scale. As can be seen in Figure 3b, Llama 2 Chat is much more consistent across samples than Llama 2 Base,

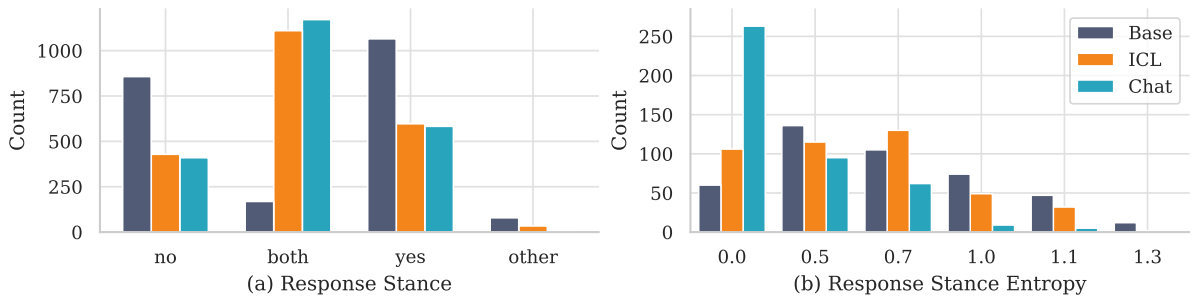

Figure 3: **(a)** Response stance distribution on the CONFLICTINGQA dataset for Llama 2 models. Aligned models provide more comprehensive responses (Overton pluaralistic, *both*) than the base model, which mostly contains one perspective (*yes/no*). **(b)** Response stance entropy distribution. Aligned models also have higher consistency within samples (low entropy).

and aligning Llama 2 Base with in-context learning (ICL) results in increased consistency.

Taken together, we observe that alignment results in reduced distributional pluralism, as most responses are "both" and samples are more homogeneous. However, the diversity of information presented in single responses is enhanced, resulting in increased Overton pluralism.

4 Mimicking aligned models

Our previous analysis showed that alignment emphasizes useful knowledge from base models and unifies more content in a single response. We now ask whether the alignment process is introducing a substantial change over the base model. We argue that if a specific capacity can be consistently evoked using a natural language prompt, it is reasonable to say that the capacity exists in the model. To investigate this, we explore more sophisticated prompting strategies to see if a base LLM aligned with ICL can match the behavior of alignment-tuned LLMs. As our goal is to mimic an existing model, we frame our discussion using terminology from the knowledge distillation literature (Hinton et al., 2015) and refer to the aligned model we wish to imitate as the *teacher* and the prompted base model as the *student*.

4.1 In-context distillation strategies

Let f and g be a teacher and student LLM respectively. Given a set of queries on which models will be evaluated, $X = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$, our goal is to construct a few-shot prompt creation function ϕ such that $f(\mathbf{x}_i) \approx g(\phi(\mathbf{x}_i))$. To construct f, we assume access to both X (our prompt dataset) as well as a separate input corpus $U = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$, a fixed set of example queries that can be used in prompts as well, and the ability to compute outputs of f on both X and U. Examples of our prompts are provided in Appendix C.

Static Prompts These prompts are static, applied globally for all input $\mathbf{x}_i$.

URIAL: Human

$$\phi(\mathbf{x}_i) = [\mathbf{u}_a; \mathbf{y}_a; \mathbf{u}_b; \mathbf{y}_b; \mathbf{u}_b; \mathbf{y}_c; \mathbf{x}_i]$$

Each query $\mathbf{u}_a, \mathbf{u}_b, \mathbf{u}_c$ is manually selected and paired with a human written response $\mathbf{y}_a, \mathbf{y}_b, \mathbf{y}_c$ respectively.

URIAL: Teacher

$$\phi(\mathbf{x}_i) = [\mathbf{u}_a; f(\mathbf{u}_a); \mathbf{u}_b; f(\mathbf{u}_b); \mathbf{u}_c; f(\mathbf{u}_c); \mathbf{x}_i]$$

This is equivalent to URIAL, except that we use responses from the teacher model instead of human-written responses.

Random: Teacher

$$\phi(\mathbf{x}_i) = [\hat{\mathbf{x}}_1; f(\hat{\mathbf{x}}_1); \hat{\mathbf{x}}_2; f(\hat{\mathbf{x}}_2); \hat{\mathbf{x}}_3; f(\hat{\mathbf{x}}_3); \mathbf{x}_i]$$

Each query $\hat{\mathbf{x}}_j \neq \mathbf{x}_i$ is randomly selected from X and paired with teacher model responses.

Dynamic Prompts These change per input $\mathbf{x}_i$.

*k*NN: Teacher

$$\phi(\mathbf{x}_i) = [\hat{\mathbf{x}}_1; f(\hat{\mathbf{x}}_1); \hat{\mathbf{x}}_2; f(\hat{\mathbf{x}}_2); \hat{\mathbf{x}}_3; f(\hat{\mathbf{x}}_3); \mathbf{x}_i]$$

Each query $\hat{\mathbf{x}}_j \in X$, $\hat{\mathbf{x}}_i \neq \mathbf{x}_i$, consists of the jth most similar queries to $\mathbf{x}_i$. To rank similarity, we use the Mistral (Jiang et al., 2023) based embedding model (Wang et al., 2023).

Oracle Prompts These prompts have access to the teacher model output $f(\mathbf{x}_i)$ for the input $\mathbf{x}_i$.

Oracle *k*NN: Teacher

$$\phi(\mathbf{x}_i) = [\hat{\mathbf{x}}_1; f(\hat{\mathbf{x}}_1); \hat{\mathbf{x}}_2; f(\hat{\mathbf{x}}_2); \hat{\mathbf{x}}_3; f(\hat{\mathbf{x}}_3); \mathbf{x}_i]$$

Like *k*NN prompts, we dynamically select $k = 3$ queries from X. However, in this setting, we measure similarity based on the teacher's response to the test query, $\mathbf{y}_i \sim f(\mathbf{x}_i)$, and the teacher's

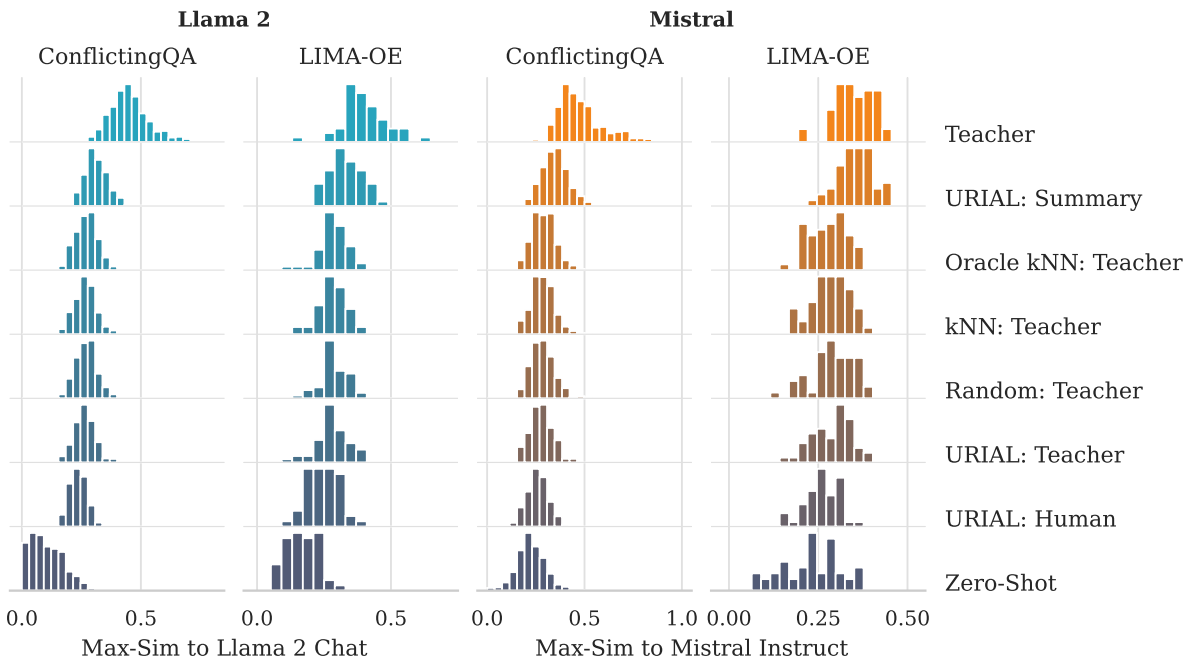

Figure 4: Histogram of maximum lexical similarity between responses for RLHFed model outputs and base model output with various ICL alignment techniques (left Llama 2, right Mistral). The top row depicts Self-Sim (max) for teacher model responses, which are fairly self-similar. Including additional context in the form of teacher responses and in-domain questions increases similarity to the teacher model, finally reaching a point with substantial distributional overlap.

response to each evaluation query, $\hat{\mathbf{y}}_j \sim f(\hat{\mathbf{x}}_j)$, $\hat{\mathbf{x}}_j \in X \setminus \{\mathbf{x}_i\}$. We use the same embedding model as *k*NN Prompts to measure similarity and adapt the embedding prompt to "retrieve similar answers" in kind.

URIAL: Summary

$$\phi(\mathbf{x}_i) = [\mathbf{u}_a; \mathbf{s}_a; f(\mathbf{u}_a); \mathbf{u}_b; \mathbf{s}_b; f(\mathbf{u}_b); \mathbf{u}_c; \mathbf{s}_c; f(\mathbf{u}_c); \mathbf{x}_i; \mathbf{s}_i]$$

Each query $\mathbf{u}_j \in U$ from the example set is paired with a response sampled from the teacher, $\mathbf{y}_j \sim f(\mathbf{u}_j)$ in addition to a summary of the response, $\mathbf{s}_j = \text{Summarize}(\mathbf{y}_j)$. Like Oracle *k*NN, this approach leverages knowledge about the teacher's response $\mathbf{y}_i$ for query $\mathbf{x}_i$ when constructing prompts, but in a more direct way as the $\mathbf{s}_j$ are included in the prompt. Prompt C.3 lists the version of this prompt with the Llama 2 Chat teacher. We use GPT-4 to summarize responses, Prompt C.7.

4.2 Evaluation criteria

Our goal is to measure whether a prompted base LLM can mimic an aligned LLM. In general, we expect even a prompted base LLM to exhibit a wider range of behaviors than an aligned LLM, but our question is whether *some* samples of that base LLM will be close to the aligned model's behavior. We therefore compute the maximum similarity between multiple samples from the base model and an aligned output $\mathbf{y}^*$.

As we will not reproduce $\mathbf{y}^*$ exactly, we characterize similarity values based on the *self*-similarity of the aligned model with itself, as defined in Section 2.3. Our rationale is that multiple aligned responses are likely to be only mildly different in terms of content, so if we can match this self-similarity with our best base model samples, we have evidence in favor of our hypothesis that the base model can get "close enough" to the aligned model's distribution.

We rely on lexical measures of response similarity. Let $\mathbf{x} \in X$ be a query, $\mathbf{y} \sim f(\mathbf{x})$ a teacher response, and $Y = \{\mathbf{y}_1, \ldots, \mathbf{y}_N\}$, $\mathbf{y}_i \sim g(\phi(\mathbf{x}))$, a set of student responses. We measure the Jaccard similarity between word stems for each $\mathbf{y}_i \in Y$ for $N = 5$ student responses and a single teacher response $\mathbf{y} \in \mathbb{V}^*$. Results are aggregated by taking the max or mean over N samples, then averaging similarities over each query $\mathbf{x} \in X$. We refer to this metric as Max-Sim.

4.3 Results

Results are presented in Figure 4. Including teacher demonstrations (URIAL/Teacher and above) leads

to significantly closer responses to the teacher response in both datasets and on both models. Dynamic prompting shows further increase in maximum similarity to teacher responses. The most similarity between student and teacher models is achieved when students are provided semantic hints about the content of teacher responses in the form of summaries (kNN Prompts and Summary Input). This is nearly equivalent to the average Self-Sim metric of the teacher, Llama 2 Chat, on LIMA-OE (0.33 vs 0.34) and only slightly smaller on CONFLICTINGQA (0.31 vs 0.36). Higher lexical similarity to aligned model responses also leads to improvement in quality metrics presented in Section 2.3. We include these results in the Appendix in Table 3.

4.4 Analysis

Qualitative example Table 5 shows responses from the zero-shot, URIAL/Human, and *k*NN/Teacher prompts compared to Llama 2 Chat. The lexical similarity of 0.45 between the teacher model and the RLHF response corresponds to a close similarity in terms of both content and formatting, while the samples with lower similarity are less similar. This suggests that similarities in the range we observe in Figure 4 are sufficient to support having "mimicked" the response. Appendix D provides more examples.

Estimate of similarity To further establish that our simple lexical similarity captures semantics of responses, we devised a rating scheme based on our observations from Table 5. Responses are rated from 0 to 2, with 0 for little match, 1 for content match but diverging format, and 2 for both format and content match. We prompted GPT-4 to assess response similarities ("Assessing Similarity" prompt in Appendix, Prompt C.8). On a set of 30 prompts each, Llama 2 responses from the *k*NN Prompts and Summary Input method were rated an average of 1.40 on CONFLICTINGQA and an average of 1.73 on LIMA-OE. This aligns with the patterns from Table 5, where the gap between teacher and *k*NN Prompts and Summary Input is larger for Conflicting QA compared to LIMA-OE. This also agreed with our manual inspection: we found that responses on LIMA-OE very frequently matched the style and content of the chat responses. On CONFLICTINGQA, there were a higher number of times when the response deviated from the expected style.

5 Related Work

Post-alignment diversity reductions Several previous works (Florian et al., 2024) have studied various types of post-alignment diversity reductions, which we highlight here. From the perspective of content diversity, prior work (Liang et al., 2023) show aligned LLMs are poorly calibrated in comparison to base LLMs. It has been shown that SFT reduces diversity less than RLHF (Kirk et al., 2024). Prior work (Lahoti et al., 2023) show models generate biased lists of people and cities post-alignment, but also find prompting can reduce this bias.

Another line of work studies reduced diversity in answers to subjective questions related to identify (Santurkar et al., 2023; Jiang et al., 2024; Sorensen et al., 2024). This is a different notion of diversity that we do not explore here and it is unclear how such findings would impact the task-oriented settings we study (Berglund et al., 2023). Prior work (Hayati et al., 2023) show LLMs can generate diverse opinions when prompted to do so and note that *"semantic diversity is not always positively correlated with perspective diversity"*, in line with the conclusions we draw on this topic.

Superficial alignment Our work is influenced by the conjecture that alignment is largely superficial (Zhou et al., 2024). Empirical evidence for the Superficial Alignment Hypothesis includes the ability to achieve comparable performance to more complex and data-intensive fine-tuning procedures using a small amount of instruction tuning data (Zhou et al., 2024; Zhao et al., 2024), in-context learning (Han, 2023; Lin et al., 2024), or proxy-tuning (Liu et al., 2024). In contrast to this prior work, which primarily focuses on showing base LLMs can match the performance of alignment-tuned LLMs according to quantitative measures of response quality, our work provides a detailed analysis of how alignment alters LLM behavior independently of any specific alignment technique.

6 Conclusion

In this paper, we investigated the relationship between LLMs before and after alignment. We revisited previous findings related to post-alignment diversity reduction in the context of open-ended QA and showed that a loss in lexical diversity (distributional pluralism) is compensated for by longer